ECEN-662: Homework 4

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Problem 1

Let $\theta > 0$ be an unknown parameter, and let $X = (X_0, X_1, \ldots, X_n)$ where $X_0 \sim Poisson(\theta)$ and for any $i \in [n]$,

$$X_i | (X_0, X_1, \ldots, X_{i-1}) \sim Poisson(\theta X_{i-1}).$$

(i) Find the MLE of θ based on the observation X.

(ii) Compute $I(\theta)$ with respect to the observation X.

Solution

Problem 2

Let $X = (X_1, \dots, X_n)$, where

$$X_i \overset{iid}{\sim} f_\theta(x_i) = \frac{1}{2}\exp(-|x_i - \theta|),$$

and $\theta > 0$ is the unknown parameter. Find an MLE of θ. (Hint: Consider the cases where n is even and odd separately and find the MLE in terms of the order statistics.)

Solution

Problem 3

Given a fixed known integer $r > 1$, let $X = X_{i,j} : i \in [n], j \in [r]$ such that

$$X_{i,j} \sim \mathcal{N}(\mu_i, \sigma^2)$$

and $X_{i,j}$'s are mutually independent. The unknown parameters are $\theta = (\mu_1, \ldots, \mu_n, \sigma^2)$, where μ_i can be any real number for all $i \in [n]$ and $\sigma^2 > 0$.

(i) Find the MLE of θ.

(ii) Is the MLE asymptotically unbiased for σ^2 as $n \to \infty$? Can you explain, intuitively, why or why not?

Solution

Problem 4

Let $X = (X_1, \ldots, X_n)$, where

$$X_i \stackrel{iid}{\sim} \theta f_1(x_i) + (1-\theta) f_2(x_i).$$

Here, f_1 and f_2 are two different known density functions, and $\theta \in [0,1]$ is the unknown parameter.

(i) Find a sufficient and necessary condition for which the score equation

$$\frac{d}{d\theta} \log f_\theta(x) = 0$$

has a unique solution. Show that if the score equation has a unique solution, it is the MLE.

(ii) Find the MLE when the score equation has no solution.

Solution

Problem 5

You have two biased coins (Coin A and Coin B) with unknown probabilities of landing on heads (θ_A and θ_B). You pick one coin at random, flip it 10 times, and record the results (e.g., (5H, 5T)). You repeat this five times for a total of 50 flips. You do not know which coin was used for each set of 10 flips. This "which coin" information is the latent variable ($U_i \in \{A, B\}, i = 1, 2, 3, 4, 5$). The goal here is to use the EM algorithm to estimate the true bias (θ_A and θ_B) for each coin.

(i) Find an explicit expression for the posterior distribution and the expectation

$$q^{(t)}(u) = f_{\theta^{(t-1)}}(u|x) \quad \text{and} \quad \mathbb{E}_{q^{(t)}}[\log f_\theta(x, U)]$$

in the E-step and the update rule

$$\theta^{(t)} = \arg\max_\theta \mathbb{E}_{q^{(t)}}[\log f_\theta(x, U)]$$

in the M-step.

(ii) Given that the observations for the five trials are:

Trial 1 (5H, 5T), Trial 2 (9H, 1T), Trial 3 (8H, 2T), Trial 4 (4H, 6T), Trial 5 (7H, 3T),

write a compute program to find the MLE of $\theta = (\theta_A, \theta_B)$. Plot the posterior probabilities $p_{\theta^{(t-1)}}(u_i = A|x)$, $i = 1, 2, 3, 4, 5$ and the estimates $(\theta_A^{(t)}, \theta_B^{(t)})$ as a function of the number of iterations t.

Solution